MAM · Rectilinear Motion — Patterns and Solution Approaches

Source: marking keys for 13 past rectilinear motion questions from 2016–2025 + exam reports (the 2018, 2024 and 2025 reports explicitly list rectilinear motion as a weakness; the 2021 report reminds students to distinguish between distance and displacement; the 2023 report specifically mentions signed area and dt notation). Original questions are in `topics/03_Rectilinear_Motion.pdf`. Review status: has undergone one round of independent review and been revised according to feedback (see `Review_log.md`).

Concept Quick Reference (the reports say this is where concepts are most easily confused)

Quantity	Definition	Note
Displacement x(t)	position relative to the origin, can be positive or negative	'returns to the starting point' is x(t) = x(0), not v = 0
Velocity v = x′(t)	can be positive or negative (direction)	'at rest' is v = 0; 'changes direction' is v changing sign
Speed = |v|	non-negative	For maximum speed, compare |v| at the extremum points and endpoints
Acceleration a = v′(t)	can be positive or negative	Where velocity has an extremum in the interior of the interval, a = 0 must occur; the converse is not necessarily true (e.g. v = t^3 at t = 0; in a constant-velocity section a is always 0); compare the endpoints separately
Change in displacement = $\int_a^b$ v dt	signed area	'What does signed area represent?' = change in displacement from t=a to t=b
Distance travelled = $\int_a^b$ |v| dt	non-negative	must split into intervals at v = 0

Speeding up or slowing down: v and a with the same sign means speeding up; with opposite signs means slowing down. The exact wording of the 2024F-Q2 key: saying only that acceleration is negative is not a valid reason. Note: The 2018A-Q11(c) key treats "heading south and decelerating at 0.5" as a = −0.5 (by the definition above, v and a then have the same sign, so it is actually speeding up). When you encounter this kind of wording, write your interpretation clearly and state the positive direction.

Time wording: 'when it first appears' and 'initially' mean t = 0 (2024 report: many students substituted t = 1 by mistake); 'during the fifth second' means 4 ≤ t ≤ 5.

Pattern 1: Given x(t), find v, a and their meaning

Approach: differentiate to get v; differentiate again to get a; substitute the time asked; write units; when explaining, make the direction clear (what the positive direction is, e.g. north, up, to the right (in 2024F-Q2, x is the distance from the left edge of the screen, and the key writes "to the right")).

Marks awarded for: ✓ expression for v(t) ✓ substitution (including units) ✓ expression for a(t) ✓ conclusion with reason (same sign/opposite signs).

Past questions: 2016F-Q4, 2016A-Q19(a), 2022A-Q10(b), 2024F-Q2(a)(b).

Pattern 2: Given a(t) or v(t), work backwards using initial conditions

Identify: "initial velocity is 5 cm/s", "x(0) = 0", "After 2 seconds displacement is −50 cm"; find constants p, k.

Approach

1. Integrate a → v (+c) → use v(0) to determine c.
2. Integrate again v → x (+c_2) → use the known position to determine c_2.
3. When the equation contains unknown parameter p: write each of the two position conditions as an

equation, then solve simultaneously for p and the constant (2017A-Q20).

4. When integrating a trigonometric v (e.g. v = 2 sin(t/3 + π/6)), divide by the inner coefficient 1/3, i.e. multiply by 3, giving −6cos(t/3 + π/6) + C; set the calculator to radians.

Marks awarded for (2018A-Q11(a)): ✓ integrate to obtain a cos form ✓ write the constant and form an equation using x(0) = 0 ✓ find C and write x(t).

Past questions: 2017A-Q20(b), 2018A-Q11, 2019A-Q9(c), 2021A-Q17(a), 2022A-Q11(c), 2023A-Q8(e) (compare coefficients to find A, B).

Pattern 3: When it is at rest / changes direction / returns to the starting point

Approach

- At rest: solve v(t) = 0. Changes direction: v must change sign (check whether v actually changes sign either side of that point; if v only touches 0 and returns to its original sign, e.g. v = (t − 2)², the object has not changed direction).
- Returns to the starting point: solve x(t) = x(0); passes through the origin: solve x(t) = 0 (these are the same only when x(0) = 0). Take the positive root that is the 'first' one required by the question.
- When only a sign table is given (2025F-Q6): x changes sign → passes through the origin; v changes sign → stops and turns around; v and a have opposite signs → slowing down. In the explanation, state that the reason is 'the sign of v changes'.

Past questions: 2016A-Q19(a), 2017A-Q20(c), 2021A-Q17(b), 2022A-Q10(a)(c), 2023A-Q8(a), 2025F-Q6.

Pattern 4: Total distance travelled vs displacement ★ must-know

Identify: "total distance travelled" and "How far has … travelled" ask for distance travelled, whereas "Evaluate ∫v dt and explain" asks for change in displacement.

Standard procedure (total distance travelled)

1. Solve v = 0 to find all times t_1, t_2… in the interval when it is stationary.
2. Write the distance expression; either form is acceptable:
   - $|x(t_1) - x(0)| + |x(t_2) - x(t_1)| + \ldots$ (the key's method for 2024F-Q2(d));
   - $\int |v(t)|\,dt$, or split into intervals: put a negative sign in front of the section where v < 0 (or put absolute value signs around each section). In 2023A-Q8(c), v = −2t sin t is negative on (π/3, π) and positive on (π, 4π/3), so it is written as −∫{π/3}^{π} v dt + ∫{π}^{4π/3} v dt. The key awards 1 mark separately for the signs, and there is also 1 mark for including dt in every integral.
3. Calculate and include units.

Interpreting $\int_a^b v\,dt$: "the change in displacement (position) of … from t = a s to t = b s" — the start and end times must also be written (2024F-Q2(c) has 1 mark specifically for the start and end times).

Marks awarded for: ✓ find the stationary times ✓ correct split expression ✓ numerical value.

Common errors: treating x(15) − x(0) directly as distance travelled; missing an intermediate turning point when splitting intervals.

Past questions: 2016A-Q19(b)(d), 2017A-Q20(d), 2018A-Q11(c), 2021A-Q14, 2021A-Q17(c), 2023A-Q8(b)(c), 2024F-Q2(c)(d).

Pattern 5: Maximum velocity / zero acceleration

Approach: set a(t) = v′(t) = 0 to find t → substitute into v → verify the maximum: confirm using v″ < 0 (or a sign test); when the question requires using calculus, this step earns marks (2022A-Q11(b) key: confirms that v″ < 0); when asked for the maximum on an interval, also compare with the endpoints. When asked for maximum speed, also compare |v| at the endpoints (2016A-Q19(c) key: examines velocity at endpoints).

Past questions: 2016A-Q19(c), 2022A-Q11(b), 2023A-Q8(d), 2021A-Q14, 2019A-Q12(c) (constant velocity for

the first 2 minutes, a is always 0, not an extremum).

Pattern 6: Interpreting context and graphs

Identify: A graph of a(t) or v(t) is given, and the question asks "explain what is happening to the velocity in $0 < t < 8$".

How to write it: "Since $a(t) > 0$ for $0 < t < 8$, the (upward) velocity is increasing; since $a(t) < 0$ for $8 < t < 16$, the velocity is decreasing (but still positive, still moving up)." Marking requirements: ✓ refer to the graph or sign of a ✓ conclusion for the first part ✓ conclusion for the second part.

Sketching (2025F-Q6(d)): the x(t) graph must satisfy all three sets of signs for x, v and a at the same time: start at the origin, have a positive initial slope, and cross the t-axis only between $3 < t < 4$; the sign of the slope corresponds to v, and concavity corresponds to a.

Past questions: 2019A-Q9(a)(b), 2025F-Q6(d), 2022A-Q10(d) (amplitude envelope $3e^{-t} = 0.01$).